

Predictive Fibres and Critical Oseen Transfer for the Three-Dimensional Navier-Stokes Equations

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Abstract

This note develops an audited presentation-theoretic regularity program for the three-dimensional incompressible Navier-Stokes equations. The basic object is a local predictive presentation: incoming and lateral data determine admissible predictors, while the compatible suitable weak states form an invisible fibre. A putative singularity is represented by a persistent failure of causal prediction across parabolic scales.

The current result is a precise conditional reduction, not yet a non-conditional regularity theorem. The paper proves several structural ingredients and isolates the remaining analytic burdens as named source theorems. The proved components include a local critical Oseen transfer estimate in the Serrin range, relaxed minimax centres, Hilbertian innovation packing, the two-mode null calculation for the Leray-projected bilinear symbol, exact Beltrami absorption, and exact two-and-a-half-dimensional reduction. The audit now also gives a detailed source-contract layer for the quotient packet route and for the pressure/gauge channel. Certified-new packet rows, representation errors, packet escapes, packet compatibility, S4 owners, pressure envelopes, pressure representatives, pressure tail production, pressure shrink/margins, and S8/Oseen owner imports are separated so that no estimate can borrow from S1, S7, the Oseen production theorem, or the final global cost statement.

Under the remaining source modules, a bad predictive chain cannot persist: either normal innovation is square-summable by Hilbert packing, the fibre enters a regular rigid branch, or a carrier/chart/envelope/margin escape is registered. The article therefore gives a professional conditional proof architecture: all closed components are proved in detail or reduced to finite ledger identities, while the remaining blocks are explicitly named, source-scoped, and forbidden from using the desired conclusion.

1 Purpose

The incompressible Navier-Stokes equations in three space dimensions are

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0.$$

The global regularity problem remains open. Classical local theory studies suitable weak solutions through scale-invariant quantities, pressure decompositions, local energy inequalities, and epsilon-regularity criteria.

This note proposes a complementary organization. Around a putative singular point, one asks whether the core of a parabolic cylinder is causally predictable from incoming and lateral information by a strong local model. The states compatible with the same incoming observables form a fibre. A singular chain is then interpreted as a sequence of scales at which no bounded predictor captures all states in the fibre.

The presentation-theoretic language is used through analytic statements. The paper separates three levels:

- (i) statements proved here;
- (ii) conditional propositions proved from explicitly stated analytic modules;
- (iii) analytic modules used as hypotheses in the modular theorem.

The level of each statement is marked explicitly by its environment: lemma, proposition, theorem, or assumption. The current manuscript is written as an audited reduction. It is meant to preserve the useful structure of the argument while making clear which statements are proved, which are formal bookkeeping closures, and which remain genuine Navier-Stokes source theorems.

2 Local Setting

2.1 Parabolic scaling

For $z_0 = (x_0, t_0)$, write

$$Q_R(z_0) = B_R(x_0) \times (t_0 - R^2, t_0).$$

The Navier-Stokes scaling is

$$u^{(R)}(x, t) = Ru(x_0 + Rx, t_0 + R^2t), \quad p^{(R)}(x, t) = R^2p(x_0 + Rx, t_0 + R^2t).$$

A quantity is critical if its normalized value is stable under this scaling.

2.2 Suitable weak solutions

A pair (u, p) is a suitable weak solution on Q if

$$u \in L_t^\infty L_x^2(Q) \cap L_t^2 \dot{H}_x^1(Q), \quad p \in L_{t,x}^{3/2}(Q),$$

the equations hold distributionally, and the local energy inequality holds. For every nonnegative $\phi \in C_c^\infty(Q)$,

$$\begin{aligned} & \int |u(x, t)|^2 \phi(x, t) dx + 2 \int_{-\infty}^t \int |\nabla u|^2 \phi dx ds \\ & \leq \int_{-\infty}^t \int |u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi dx ds. \end{aligned}$$

2.3 Pressure cleaning

Let $B \subset \mathbb{R}^3$ be a smooth ball. For $1 < s < \infty$, the local Helmholtz projection $\mathbb{P}_B$ is

$$\mathbb{P}_B f = f - \nabla \pi_B,$$

where π_B solves the Neumann problem

$$\Delta \pi_B = \nabla \cdot f, \quad \partial_n \pi_B = f \cdot n,$$

with zero average normalization. Standard elliptic estimates give

$$\|\mathbb{P}_B f\|_{L^s(B)} + \|\nabla \pi_B\|_{L^s(B)} \leq C_{s,B} \|f\|_{L^s(B)}.$$

For a local argument, the pressure is decomposed schematically as

$$p = p_{\text{loc}} + p_{\text{harm}} + p_{\text{tail}}.$$

The local part is absorbed by $\mathbb{P}_B$. The harmonic and nonlocal pieces are not ignored: their contribution is registered by a nonnegative quantity

$$\text{PressTail}(Q_R, Q_r).$$

In each estimate below, this term means precisely the residual pressure contribution not controlled by the local projection.

3 Predictive Presentations

3.1 Incoming data

Fix $0 < \theta < 1$ and $0 < \lambda < 1$, with $\theta^2 < \lambda$. For $Q_R(z_0)$, define the incoming region

$$\mathcal{I}^-(Q_R) = B_R(x_0) \times (t_0 - R^2, t_0 - \lambda R^2) \cup (B_R(x_0) \setminus B_{\theta R}(x_0)) \times (t_0 - R^2, t_0).$$

It contains a past slab and a lateral collar, but excludes the final core.

3.2 Predictors and fibres

A causal strong predictor on Q_R is a strong solution g constructed from data measurable on $\mathcal{I}^-(Q_R)$, after the chosen pressure-cleaning procedure. The critical predictor budget is

$$\|g\|_{L_t^q L_x^p(Q_R)} \leq L, \quad \frac{2}{q} + \frac{3}{p} = 1, \quad p > 3.$$

Additional budget components may record pressure-tail size, observable cost, localization cost, and conditioning of the finite observational Gramian.

Let $\mathcal{O}$ be a finite family of incoming observables. The invisible fibre associated with (u, p) is the class

$$\mathcal{S}_{L, \mathcal{O}}^-(Q_R)$$

of suitable weak states (v, q) in Q_R with budget at most L and the same incoming observations as (u, p) :

$$\mathcal{O}(v, q) = \mathcal{O}(u, p).$$

Let $\mathcal{P}_{L, \mathcal{O}}^-(Q_R)$ denote the corresponding class of causal strong predictors.

Definition 3.1 (Local predictive defect). *Let d_{Q_R} be a pressure-cleaned distance on the core $Q_{\theta R}$. The minimax predictive defect is*

$$\mathfrak{D}_{L, \mathcal{O}}(Q_R) = \inf_{g \in \mathcal{P}_{L, \mathcal{O}}^-(Q_R)} \sup_{(v, q) \in \mathcal{S}_{L, \mathcal{O}}^-(Q_R)} d_{Q_R}((v, q), g).$$

When the infimum is taken in the closed convex hull of the predictor class, the distance from the relaxed centre to realizable predictors is recorded as a realization defect $\mathfrak{V}(Q_R)$.

A bad predictive chain at z_0 is a nested sequence $Q_{R_k}(z_0)$, $R_k \downarrow 0$, for which $\mathfrak{D}_{L, \mathcal{O}_k}(Q_{R_k})$ stays bounded below by a fixed positive number.

4 Critical Local Oseen Transfer

The following estimate is the most classical analytic ingredient of the program. It is included because it is the mechanism that transports a perturbation around a Serrin-critical strong predictor from one scale to a smaller core.

Lemma 4.1 (Critical local Oseen transfer). *Let $Q_r \Subset Q_R$. Let g be divergence-free and*

$$g \in L_t^q L_x^p(Q_R), \quad \frac{2}{q} + \frac{3}{p} = 1, \quad p > 3.$$

Let r be divergence-free and pressure-cleaned, satisfying

$$\partial_t r - \Delta r + (g \cdot \nabla) r + (r \cdot \nabla) g + \nabla q = F$$

in Q_R , distributionally. Choose $\zeta \in C_c^\infty(Q_R)$ with $\zeta = 1$ on Q_r . Assume that the nonlocal pressure contribution is bounded by $\text{PressTail}_{R,r}$, and define the incoming-collar cost by

$$\begin{aligned} \text{Coll}_{\zeta,g}(r)^2 &= \sup_{\text{initial slab}} \int |r|^2 \zeta^2 dx \\ &\quad + \int_{Q_R} |r|^2 \left(|\partial_t \zeta| + |\Delta \zeta| + |\nabla \zeta|^2 \right) dx dt \\ &\quad + \int_{Q_R} |g| |r|^2 |\nabla \zeta| dx dt. \end{aligned}$$

Then

$$\begin{aligned} &\|r\|_{L_t^\infty L_x^2(Q_r)} + \|\nabla r\|_{L_{t,x}^2(Q_r)} + \|r\|_{L_{t,x}^{10/3}(Q_r)} \\ &\leq C \left(\text{Coll}_{\zeta,g}(r) + \|F\|_{L_t^2 H_x^{-1}(Q_R)} + \text{PressTail}_{R,r} \right), \end{aligned}$$

where C depends on $Q_r \Subset Q_R$, on p , and on $\|g\|_{L_t^q L_x^p(Q_R)}$.

Proof. Test the equation against $r\zeta^2$. The local pressure projection removes the local pressure contribution; the remaining pressure term is bounded by the definition of $\text{PressTail}_{R,r}$. We obtain

$$\frac{1}{2} \frac{d}{dt} \int |r|^2 \zeta^2 + \int |\nabla r|^2 \zeta^2 \leq I_{\text{cut}} + I_g + I_F + I_{\text{press}}.$$

The cutoff term is bounded by the collar part of $\text{Coll}_{\zeta,g}(r)^2$. The forcing term is estimated by duality:

$$|\langle F, r\zeta^2 \rangle| \leq \|F\|_{H^{-1}} \|r\zeta^2\|_{H^1} \leq \frac{1}{4} \|\zeta \nabla r\|_2^2 + C \|F\|_{H^{-1}}^2 + C \|r \nabla \zeta\|_2^2.$$

Since $\nabla \cdot g = 0$,

$$\int (g \cdot \nabla r) \cdot r\zeta^2 = -\frac{1}{2} \int |r|^2 g \cdot \nabla(\zeta^2),$$

which is included in the collar cost.

It remains to estimate the stretching term. Using $\nabla \cdot r = 0$,

$$\int r_i (\partial_i g_j) r_j \zeta^2 = - \int g_j r_i (\partial_i r_j) \zeta^2 - \int g_j r_i r_j \partial_i (\zeta^2).$$

The second term is again a collar term. For the first,

$$\left| \int g_j r_i (\partial_i r_j) \zeta^2 \right| \leq \|g\|_{L^p} \|r\zeta\|_{L^{2p/(p-2)}} \|\zeta \nabla r\|_{L^2}.$$

By Gagliardo-Nirenberg,

$$\|r\zeta\|_{L^{2p/(p-2)}} \leq C \|r\zeta\|_{L^2}^{1-3/p} \|\nabla(r\zeta)\|_{L^2}^{3/p}.$$

Writing

$$E(t) = \int |r|^2 \zeta^2, \quad D(t) = \int |\nabla r|^2 \zeta^2,$$

and absorbing the cutoff contribution into the collar cost, Young's inequality gives

$$|I_g^{\text{int}}| \leq \frac{1}{4} D(t) + C \|g(t)\|_{L^p}^q E(t), \quad q = \frac{2p}{p-3}.$$

The exponent is exactly the Serrin exponent because $2/q + 3/p = 1$.

Thus

$$\frac{d}{dt} E(t) + \frac{1}{2} D(t) \leq C \|g(t)\|_{L^p}^q E(t) + B(t),$$

where $B(t)$ contains only forcing, pressure-tail, and collar terms. Since $\|g(t)\|_{L^p}^q \in L_t^1$, Gronwall yields

$$\sup_t E(t) + \int D(t) dt \leq C \left(\text{Coll}_{\zeta, g}(r)^2 + \|F\|_{L_t^2 H_x^{-1}(Q_R)}^2 + \text{PressTail}_{R, r}^2 \right).$$

Finally,

$$\|r\|_{L^{10/3}(Q_r)} \lesssim \|r\|_{L_t^\infty L_x^2(Q_r)}^{2/5} \|\nabla r\|_{L^2(Q_r)}^{3/5}.$$

This proves the claim. $\square$

5 Minimax Centres

The minimax construction used by the predictive framework is a Hilbert-space statement. The PDE content lies in verifying that the chosen predictor classes satisfy the hypotheses.

Lemma 5.1 (Relaxed minimax centres). *Let $\mathcal{H}$ be a Hilbert space. Let $P \subset \mathcal{H}$ be nonempty, bounded, closed, and convex, and let $S \subset \mathcal{H}$ be bounded. Define*

$$R(g) = \sup_{v \in S} \|v - g\|_{\mathcal{H}}, \quad g \in P.$$

Then R attains its minimum on P .

Proof. The map $g \mapsto \|v - g\|_{\mathcal{H}}$ is convex and weakly lower semicontinuous for each v . Hence R , as a supremum of weakly lower semicontinuous functions, is weakly lower semicontinuous. Since Hilbert spaces are reflexive, a bounded closed convex subset is weakly compact. Therefore R attains its minimum on P . $\square$

Remark 5.2. *In the Navier-Stokes application, P is the weakly closed convex hull of admissible strong predictors at a fixed finite observation budget. The gap between a relaxed minimax centre and an actual strong predictor is exactly the realization defect $\mathfrak{V}$. This defect must be paid for in the scale-transfer inequality.*

6 Innovation Packing

6.1 Abstract packing

Let $\mathfrak{H}$ be a Hilbert space of normalized adjoint observables. Suppose that a sequence of normal defects $\mathcal{N}_k \in \mathfrak{H}^*$ has normalized innovative witnesses $e_k \in \mathfrak{H}$. The filtration condition is

$$e_k \perp \mathcal{P}_{k-1}, \quad \mathcal{P}_k = \overline{\mathcal{P}_{k-1} + \text{span}\{e_k\}},$$

so (e_k) is orthonormal on innovative scales.

Lemma 6.1 (Hilbertian innovation packing). *Assume that there exist $\mathcal{R} \in \mathfrak{H}^*$, errors ρ_k , and $c > 0$ such that*

$$\langle \mathcal{N}_k, e_k \rangle = \langle \mathcal{R}, e_k \rangle + \rho_k, \quad \sum_k |\rho_k|^2 < \infty,$$

and

$$|\langle \mathcal{N}_k, e_k \rangle| \geq c \|\mathcal{N}_k\|_{\mathfrak{H}^*}.$$

Then

$$\sum_k \|\mathcal{N}_k\|_{\mathfrak{H}^*}^2 \leq C_c \left(\|\mathcal{R}\|_{\mathfrak{H}^*}^2 + \sum_k |\rho_k|^2 \right).$$

Proof. By Bessel's inequality,

$$\sum_k |\langle \mathcal{R}, e_k \rangle|^2 \leq \|\mathcal{R}\|_{\mathfrak{H}^*}^2.$$

Using the coefficient representation,

$$\sum_k |\langle \mathcal{N}_k, e_k \rangle|^2 \leq 2 \sum_k |\langle \mathcal{R}, e_k \rangle|^2 + 2 \sum_k |\rho_k|^2.$$

The witness lower bound then gives the stated estimate. $\square$

6.2 PDE meaning

The lemma is abstractly complete. Its PDE use is the global residual representation module: all normalized defects from all scales must be transported into one Hilbert space with square-summable connection, cutoff, gauge, and pressure errors.

7 Repaired Cost and Quotient Packet Source Contracts

The audit of the proof is most stringent at the point where Hilbert innovation is converted into packet estimates. Hilbert orthogonality of the innovation witnesses does not by itself give a Navier-Stokes packet row estimate. The row family must be realized in a fixed or finitely merged packet atlas, and every error surrounding the row estimate must have an owner independent of the row estimate itself.

7.1 Finite repaired source ledger

The repaired cost is organized as a finite weighted source expansion

$$Q_k^{\text{rep}} = \text{Innov}_k + \sum_{s \in S_{\text{err}}} w_s \text{Err}_{s,k} + \sum_{x \in S_{\text{esc}}} w_x \text{Esc}_{x,k} + \sum_{u \in S_{\text{upd}}} w_u \text{Upd}_{u,k}.$$

For each source channel exactly one of the following outcomes is legal:

- (i) a summable source theorem;
- (ii) a finite update or finite-atlas merge with registered boundary cost;
- (iii) a named non-summable escape;
- (iv) a gate, meaning entry into a regularity or predictive closure alternative;
- (v) an explicitly open PDE theorem.

Only the first two outcomes contribute finite error terms on a no-gate tail. A named non-summable escape exits the no-escape theorem; it is not a finite error in disguise. An explicitly open PDE theorem records an honest remaining analytic burden and cannot be used as a closed input to the final proof.

7.2 The S1 quotient packet row

For an innovative scale, let

$$e_k = \frac{d_k - P_{V_{k-1}} d_k}{\|d_k - P_{V_{k-1}} d_k\|}, \quad m_k = A_k^* e_k.$$

In a fixed packet frame write

$$m_k = \text{Syn}_E(a_k) + \tau_k, \quad a_k = a_k^{\text{old}} + a_k^{\text{cert}} + a_k^{\text{new}} + a_k^{\text{tail}} + a_k^{\text{esc}}.$$

The S1 row theorem is reserved for the certified-new part a_k^{new} . Old, certified, recurrent, tail, and escape components are routed to their own owners before S1 is invoked.

Assumption 7.1 (S1 quotient packet row source). *On a fixed or finitely merged packet tail, the certified-new rows satisfy one of the standard fixed-atlas row criteria: finite overlap, Schur/Gram summability, a Carleson packet estimate, or a direct row-operator bound. Equivalently, the operator*

$$T_{\text{new}c} = (\langle c, a_k^{\text{new}} \rangle)_k$$

is bounded on the fixed packet coefficient space. The allowed conclusion is

$$\sum_k \|a_k^{\text{new}}\|_{\ell^2}^2 \leq C_{S1} + C_{S1} \sum_k \text{ErrPktCompat}_k^{\text{ind}} + C_{S1} \sum_k \text{EscPkt}_k,$$

where the two right-hand side channels have source outcomes independent of this row theorem.

Remark 7.2. *This assumption is still one of the central Navier-Stokes PDE tasks. It may not be inferred from Hilbert Gram-Schmidt orthogonality of the e_k 's, because the rows a_k^{new} live in scale-dependent packet coordinates with pressure, cutoff, carrier, gauge, and chart transport errors.*

7.3 Representation inequality

Let $b_k = \text{An}_E(d_k)$ be the fixed-frame coefficient vector of the registered defect. The packet pairing contract requires

$$\langle d_k, e_k \rangle_{H_{\text{def}}} = \langle b_k, a_k \rangle_{\ell^2} + \rho_{\text{frame},k} + \rho_{\text{reg},k} + \rho_{\text{native},k}.$$

All discrepancies are owned by the repaired ledger. If

$$\|b_k\|_{\ell^2} \leq M_{\text{pkt}},$$

then Cauchy-Schwarz and Young give

$$\text{Innov}_k \leq C_{\text{repr}} \|a_k^{\text{new}}\|_{\ell^2}^2 + C_{\text{repr}} \text{ErrS1repr}_k^{\text{ind}}.$$

The independent representation error is decomposed as

$$\begin{aligned} \text{ErrS1repr}_k^{\text{ind}} &= \text{ErrPairFrame}_k + \text{ErrPairReg}_k + \text{ErrPairNative}_k \\ &\quad + \text{ErrOldPair}_k + \text{ErrCertPair}_k + \text{ErrTailPair}_k + \text{ErrPairEnv}_k. \end{aligned}$$

The seven owners are, respectively, the packet-frame owner, the registration owner, the native component owners, the memory/span-gap owner, the certified physical subtraction owner, the tail owner, and the packet-envelope owner.

Proposition 7.3 (S1 representation and escape source contract). *The representation inequality is a closed input to S1 only after every component of $\text{ErrS1repr}^{\text{ind}}$ has a legal source outcome independent of S1. Packet escape is legal on the S1 right-hand side only in the registered summable case. More precisely,*

$$\begin{aligned} \text{EscPkt}_k &= \text{EscFrame}_k + \text{EscCarrier}_k + \text{EscDict}_k + \text{EscEnvelope}_k \\ &\quad + \text{EscOwnerSplit}_k + \text{EscRecurrent}_k + \text{EscMarginFloor}_k + \text{EscAtlasVar}_k. \end{aligned}$$

Each packet escape component must be registered summable, finitely updated or merged, a named non-summable exit, a gate, or an explicitly open PDE theorem. A large value of $\|a_k^{\text{new}}\|_{\ell^2}^2$ alone is not a packet escape.

Proof. This is a bookkeeping theorem in the finite repaired ledger. The representation terms are precisely the discrepancies between the Hilbert pairing and the fixed packet pairing, plus old/certified/tail/envelope pairings. By construction none of these terms contains the certified-new row norm, and each has a primary owner in S2-S12, S4, S5, the packet-envelope ledger, finite variation, or a named escape. The escape list records failures of packet data, not large row mass. Therefore the row estimate and the representation inequality can use these terms without proving their own right-hand sides. $\square$

7.4 S4 packet compatibility owner matrix

The packet compatibility error that may appear beside S1 is the independent part with the certified-new row mass removed:

$$\text{ErrPktCompat}_k = \text{ErrPktCompat}_k^{\text{ind}}.$$

It is decomposed as

$$\begin{aligned} \text{ErrPktCompat}_k^{\text{ind}} = & \text{FrameCompat}_k + \text{RowTailCompat}_k + \text{SectorCompat}_k + \text{DictCompat}_k \\ & + \text{OwnerSplitCompat}_k + \text{CertOldCompat}_k \\ & + \text{EnvCompat}_k + \text{MarginPktCompat}_k. \end{aligned}$$

The eight primary owner ledgers are:

$$R_{\text{frame}}, \quad R_{\text{tail}}, \quad R_{\text{sector}}, \quad R_{\text{dict}}, \quad R_{\text{owner}}, \quad R_{\text{cert}}, \quad R_{\text{env}}, \quad R_{\text{margin}}.$$

The local S4 estimate on finite legal blocks is

$$\begin{aligned} \sum_{k \in B} \text{ErrPktCompat}_k^{\text{ind}} \leq C_{S4} \sum_{k \in B} & (R_{\text{frame},k} + R_{\text{tail},k} + R_{\text{sector},k} + R_{\text{dict},k} \\ & + R_{\text{owner},k} + R_{\text{cert},k} + R_{\text{env},k} + R_{\text{margin},k}). \end{aligned}$$

Proposition 7.4 (S4 owner aggregation firewall). *The aggregate symbol R_{S4} is only shorthand for the eight owner ledgers above. It is not an independent source theorem. On an infinite no-gate no-escape tail, S4 summability follows only if all eight owner families have summable source outcomes after finitely many certified updates or merges. If one owner exits by named escape or gate, the S4 summability route exits; if one owner is only an open PDE theorem, S4 remains conditional.*

Proof. The finite-block estimate is the finite sum of the eight local owner estimates. Passing to an infinite tail is justified only by monotone convergence after each owner tail has a finite source theorem and all update boundary costs are registered. There is no ninth analytic estimate hidden in the notation R_{S4} . $\square$

7.5 Current packet-accounting status

The packet-accounting layer is therefore formally closed in the following sense:

- (i) the certified-new row is the only object counted by S1;
- (ii) the representation error has seven independent owners;
- (iii) packet escape has eight named components and cannot be large new-row mass;
- (iv) packet compatibility has eight S4 owners;

- (v) S4 aggregation is a finite owner sum, not a separate source theorem;
- (vi) no S1 surrounding source may use S1 row Bessel, final GQ1, or a threshold chosen after the row is known.

The analytic burden that remains is correspondingly sharp: prove the actual source outcomes for the representation owners, packet escapes, and eight S4 owner families; prove the true S1 row Bessel/Carleson theorem for Navier-Stokes packet rows; and stabilize the packet atlas through the finite-variation and finite-atlas theorems.

8 The Bilinear Symbol and Null Fibres

The next lemma is a finite-dimensional calculation for the Leray-projected quadratic symbol. It explains why large invisible fibres with very small quadratic leakage should be close to special geometric branches, but by itself it treats only pure modes.

Lemma 8.1 (Two-mode null classification). *Let*

$$u = ae^{i\xi \cdot x}, \quad v = be^{i\eta \cdot x},$$

with

$$a \cdot \xi = 0, \quad b \cdot \eta = 0.$$

Set $k = \xi + \eta$, $k \neq 0$, and consider the symmetrized projected symbol

$$\mathfrak{b}((\xi, a), (\eta, b)) = \mathbb{P}_k [(a \cdot \eta)b + (b \cdot \xi)a].$$

Write

$$a = \alpha k + A, \quad b = \beta k + B, \quad A \perp k, \quad B \perp k.$$

Then $\mathfrak{b} = 0$ if and only if

$$\alpha B + \beta A = 0.$$

The nonzero solutions fall into the following branches:

- (i) *the shear-null branch $\alpha = \beta = 0$;*
- (ii) *the balanced equal-radius branch $\alpha\beta \neq 0$, which forces $|\xi| = |\eta|$.*

Proof. Since $a \cdot \xi = 0$ and $b \cdot \eta = 0$,

$$a \cdot \eta = a \cdot (\xi + \eta) = a \cdot k, \quad b \cdot \xi = b \cdot (\xi + \eta) = b \cdot k.$$

Therefore

$$(a \cdot \eta)b + (b \cdot \xi)a = (a \cdot k)b + (b \cdot k)a = |k|^2(\alpha b + \beta a).$$

Substituting the decompositions gives

$$\alpha b + \beta a = 2\alpha\beta k + \alpha B + \beta A.$$

The projection $\mathbb{P}_k$ kills the component parallel to k , so the projected symbol vanishes exactly when $\alpha B + \beta A = 0$.

If $\alpha = \beta = 0$, both polarizations are perpendicular to k , giving the shear-null branch. If exactly one of α, β is zero, then the relation forces one polarization to vanish, so there is no nonzero two-mode interaction. If $\alpha\beta \neq 0$, then $B = -(\beta/\alpha)A$. The incompressibility condition for a gives

$$A \cdot \xi = -\alpha k \cdot \xi.$$

Since $A \perp k$, we have $A \cdot \eta = -A \cdot \xi = \alpha k \cdot \xi$. The incompressibility condition for b gives

$$0 = b \cdot \eta = \beta k \cdot \eta + B \cdot \eta = \beta k \cdot \eta - \frac{\beta}{\alpha} A \cdot \eta = \beta(k \cdot \eta - k \cdot \xi).$$

Thus $k \cdot \eta = k \cdot \xi$, which is

$$|\eta|^2 + \xi \cdot \eta = |\xi|^2 + \xi \cdot \eta.$$

Hence $|\xi| = |\eta|$. □

Remark 8.2. *The next analytic step is a quantitative packet-level null-graph rigidity theorem: it should promote this exact two-mode calculation to localized parabolic wave packets and identify when a large fibre with small leakage is close to a Beltrami, shear, or carrier-escape branch.*

9 Rigid Branches

Proposition 9.1 (Beltrami absorption). *Suppose U is divergence-free and*

$$\nabla \times U = \lambda U.$$

Then

$$(U \cdot \nabla)U = \nabla \left(\frac{|U|^2}{2} \right).$$

Thus the quadratic term is pure pressure and is eliminated by Leray projection or by local pressure-cleaning.

Proof. Use the identity

$$(U \cdot \nabla)U = \nabla \left(\frac{|U|^2}{2} \right) - U \times (\nabla \times U).$$

If $\nabla \times U = \lambda U$, then $U \times (\lambda U) = 0$. □

Proposition 9.2 (Two-and-a-half-dimensional reduction). *Let*

$$U(x, t) = V(y, t) + W(y, t)n,$$

where y belongs to a fixed plane Π , $V(y, t) \in \Pi$, $n \perp \Pi$, and $\nabla_\Pi \cdot V = 0$. Then the three-dimensional Navier-Stokes system reduces to

$$\partial_t V + V \cdot \nabla_\Pi V - \Delta_\Pi V + \nabla_\Pi P = 0,$$

$$\partial_t W + V \cdot \nabla_\Pi W - \Delta_\Pi W = 0.$$

Thus the planar part is a two-dimensional Navier-Stokes solution and the transverse component is a passive transported-diffused scalar.

Proof. All fields are constant in the n -direction. The in-plane component of the equation is exactly the two-dimensional incompressible Navier-Stokes equation for V . The n -component contains no pressure gradient and gives the passive scalar equation for W . □

10 The Remaining Analytic Modules

We now state the analytic modules still needed by the closed regularity argument. The previous section fixed the source-contract bookkeeping. The assumptions below are the remaining scale-invariant PDE assertions.

Assumption 10.1 (Predictive epsilon regularity). *For every L and every admissible pressure-cleaning scheme, there exist $\varepsilon(L) > 0$ and $0 < \theta < 1$ such that the following holds. If (u, p) is suitable in Q_1 , and if there is a strong predictor g with*

$$\|g\|_{L_t^q L_x^p(Q_1)} \leq L, \quad \frac{2}{q} + \frac{3}{p} = 1, \quad p > 3,$$

such that

$$\|u - g\|_{X(Q_\theta)} + \text{PressTail}(Q_1, Q_\theta) \leq \varepsilon(L),$$

then u is regular in a smaller cylinder $Q_{\theta/2}$. Here X is the pressure-cleaned local energy-critical control norm used in the predictive distance.

Assumption 10.2 (Critical predictor-map differentiability). *The causal strong solution map Φ , from incoming data to core predictors, is twice differentiable in the selected critical topology on each bounded predictor class. Quantitatively, for incoming data a and perturbations h ,*

$$\|\Phi(a + h) - \Phi(a) - D\Phi(a)h\|_X \leq C_L \|h\|_{\mathcal{A}}^2,$$

with constants stable under parabolic rescaling and compatible with pressure-cleaning.

Assumption 10.3 (Global residual representation). *Along every bad predictive chain, there is a normalized Hilbert space $\mathfrak{H}$, quasi-isometric identifications of the scale-dependent witness spaces with $\mathfrak{H}$, a single residual $\mathcal{R} \in \mathfrak{H}^*$, and square-summable errors ρ_k such that*

$$\langle \mathcal{N}_k, e_k \rangle = \langle \mathcal{R}, e_k \rangle + \rho_k$$

for every innovative normal witness e_k . The errors include pressure tails, cutoffs, holonomy, gauge changes, and finite-window transport errors.

Assumption 10.4 (Null-graph rigidity for packets). *If a localized parabolic wave-packet family has quadratically small normal leakage on a quantitatively connected interaction graph, then, after removing pressure and gauge directions, it is close in the critical norm to one of the classified branches: Beltrami, two-and-a-half-dimensional shear, or carrier escape. If it is not close to such a branch, it produces a registered normal defect.*

Assumption 10.5 (Carrier localization and collapse). *High-frequency escape either admits a carrier recentering that returns the dominant frequency to order one, or is locally subcritical in every carrier cell, or regenerates critical mass through a registered normal, holonomy, pressure-tail, or localization defect.*

Assumption 10.6 (Quantitative closure alternative). *For every L and $\delta > 0$, there are $m = m(L, \delta)$ and $c = c(L, \delta) > 0$ such that the following holds along any m -step controlled block of a predictive chain. If $\mathfrak{D}_k \geq \delta$ throughout the block and all registered normal, holonomy, pressure, carrier, and realization defects in the block are at most c , then either the defect decreases below $\delta/2$ at the next controlled scale, or the chain enters a rigid branch or a carrier branch covered by the previous assumptions.*

Assumption 10.7 (Completion of the repaired source ledger). *Every source channel in the repaired ledger has one of the legal outcomes listed in the finite source expansion:*

- (i) a summable source theorem;
- (ii) a finite update or finite-atlas merge with registered boundary cost;
- (iii) a named non-summable escape;
- (iv) a regularity gate;
- (v) an explicitly open PDE theorem.

For the final non-conditional theorem, the fifth alternative must be eliminated. In particular, the following families require actual Navier-Stokes source theorems: the eight S_4 owner families, the seven ErrS1repr owners, the eight EscPkt components, the recurrent/Hilbert-old alternatives, the $S2$ - $S12$ transport and component ledgers, the finite-variation route VarEff , and the exact $E12$ floor.

Assumption 10.8 (Exact floor and finite-atlas closure). *The $E12$ floor used in the block selection is chosen after all relevant weights, packet envelopes, margin states, and update costs are fixed. On every no-gate no-escape tail, either the finite-atlas and finite-variation states stabilize after finitely many registered updates, or the branch exits by a named chart, packet, carrier, margin, floor, or component escape. The floor is never recomputed for an already selected block without paying an explicit transport or update cost.*

11 Conditional Newton Transfer

Under the differentiability module, the Newton bridge becomes a precise conditional proposition.

Proposition 11.1 (Conditional saddle Newton bridge). *Assume critical predictor-map differentiability and the local Oseen transfer estimate. Let g_k be a relaxed minimax predictor at scale k , and write $u = g_k + w_k$. Then, after passing to the next core scale,*

$$\mathfrak{D}_{k+1} \leq C_L \mathfrak{D}_k^2 + C_L \left(\mathfrak{R}_k^\perp + \mathfrak{H}_k^\perp + \mathfrak{F}_k^\perp + \mathfrak{V}_k \right) + \eta_k.$$

Here $\mathfrak{R}_k^\perp$ is the non-absorbed normal quadratic defect, $\mathfrak{H}_k^\perp$ the holonomy or gauge defect, $\mathfrak{F}_k^\perp$ the pressure/localization defect, $\mathfrak{V}_k$ the realization defect of the relaxed centre, and η_k the chosen minimization error.

Proof. The predictable linear part of w_k is absorbed by the first variation $D\Phi(a)h$. The tangential quadratic part is absorbed by the second-order correction supplied by the differentiability module. The non-absorbed quadratic part is exactly $\mathfrak{R}_k^\perp$; pressure, cutoff, and connection errors give $\mathfrak{F}_k^\perp$ and $\mathfrak{H}_k^\perp$; and non-realizability of the relaxed centre gives $\mathfrak{V}_k$. Comparing the suitable weak solution with the updated strong predictor, the local Oseen estimate bounds the remainder on the smaller core. Since the updated predictor is an admissible competitor for the minimax problem at the next scale, the displayed inequality follows. $\square$

12 Modular Regularity Theorem

Theorem 12.1 (Audited conditional predictive regularity). *Let (u, p) be a suitable weak solution near z_0 . Suppose that the predictive presentation system at z_0 satisfies the source contracts of the repaired ledger, the analytic modules stated above, the $S1$ quotient packet row source, the $S7$ pressure/gauge source blocks $GP1$ – $GP5$, and the exact floor/finite-atlas closure. Suppose also that the minimization errors η_k are summable and that no source channel is left in the*

explicitly open PDE category. Then there is no bad predictive chain at z_0 . Consequently, if the predictive defect is the only obstruction to predictive epsilon regularity, then u is regular at z_0 .

Proof. Assume that a bad predictive chain exists. Then for some $\delta_0 > 0$,

$$\mathfrak{D}_k \geq \delta_0$$

on all sufficiently small scales.

By the global residual representation and the Hilbertian packing lemma, the innovative normal defects are square-summable. The repaired source ledger and the S1/S4 firewalls ensure that the surrounding packet, representation, escape, transport, pressure, cutoff, carrier, margin, and floor errors are either summable on the no-gate tail, finitely updated before the next epoch, or have already forced a named escape or gate. The GP1–GP5 source blocks ensure, in particular, that pressure-envelope changes, pressure representative drift, pressure-tail production, pressure shrink/margins, and S8/Oseen pressure imports have single owners and pre-E12 ledgers before 243 and 265 are invoked. Thus on a genuine no-gate no-escape tail, blocks of arbitrarily small scales have very small average registered defect.

The quantitative closure alternative prevents a persistent defect $\mathfrak{D}_k \geq \delta_0$ on such blocks unless the chain enters a rigid branch or a carrier branch.

If the chain enters a Beltrami branch, the quadratic term is pressure-cleaned and the Newton transfer gives decay of the predictive defect. If it enters a two-and-a-half-dimensional branch, the reduced two-dimensional system and passive scalar equation provide the strong predictor and again force decay, up to a registered anisotropic defect. If the chain enters carrier escape, the carrier localization module either recenters the scale, makes the contribution locally subcritical, or produces one of the registered defects already controlled by packing.

Thus every legal alternative contradicts the persistence of $\mathfrak{D}_k \geq \delta_0$. Therefore no bad predictive chain exists. The final implication is exactly predictive epsilon regularity. $\square$

13 Verified Components and Remaining Work

The verified pieces are:

- (i) the Serrin-critical Oseen transfer estimate, with collar and pressure-tail terms explicitly paid for;
- (ii) the Hilbert-space existence of relaxed minimax centres;
- (iii) the abstract Bessel packing mechanism for innovative normal witnesses;
- (iv) the exact two-mode null classification of the projected bilinear symbol;
- (v) exact absorption of Beltrami fields and exact reduction of two-and-a-half-dimensional fields.

The formal source-contract layer now additionally closes:

- (i) the quotient packet split into old, certified, new, tail, and escape rows;
- (ii) the representation inequality $\text{Innov}_k \leq C \|a_k^{\text{new}}\|_{\ell^2}^2 + C \text{ErrS1repr}_k^{\text{ind}}$;
- (iii) the seven-owner decomposition of ErrS1repr ;
- (iv) the eight-component decomposition of EscPkt , with summable escape separated from non-summable branch exit;

- (v) the eight-owner S4 packet-compatibility matrix;
- (vi) the S4 aggregation firewall, which prevents R_{S4} from being used as an independent source theorem;
- (vii) the rule that S1, S4, E12, and final GQ1 cannot prove the source estimates that they consume.
- (viii) the S7 pressure/gauge source architecture: only spatial constants are zero-cost, nonconstant harmonic pressure transfers are pressure-force pair morphisms, and the GaugePress channel is conditionally assembled from five named source blocks GP1–GP5.

The remaining analytic work is concentrated in the following PDE source theorems:

- (i) prove a true S1 row Bessel/Carleson theorem for the certified-new Navier-Stokes packet rows;
- (ii) prove the packet pairing identity and packet-envelope outcomes for actual registered Navier-Stokes defects;
- (iii) prove source outcomes for the seven representation-error owners;
- (iv) prove source outcomes or legal exits for the eight packet-escape components;
- (v) prove source outcomes for all eight S4 owner families: frame, row-tail and native, carrier-sector, dictionary and component, owner-split, certified-old, packet-envelope, and margin/floor;
- (vi) prove recurrent and Hilbert-old alternatives through the certified and recurrent ledgers, without using S1;
- (vii) prove finite-atlas and finite-variation closure through the VarEff route;
- (viii) prove the exact E12 floor and its compatibility with packet envelopes, margins, updates, and block selection;
- (ix) prove the pressure/gauge source blocks GP1–GP5: pressure-envelope source closure, pressure representative variation, pressure-tail block production, pressure shrink/margin source closure, and S8/Oseen/S6/S11 owner-split import;
- (x) prove the shared PressureForcePair source theorem and the S8 pressure/collar and explicit Oseen-error source outcomes needed by S7;
- (xi) prove critical predictor-map differentiability, global residual representation, packet-level null-graph rigidity, carrier localization, and the quantitative closure alternative in the concrete Navier-Stokes topology.

The useful conclusion of the note is therefore a dependency graph with firewalls. Presentation theory supplies the language of predictors, observables, fibres, costs, owners, and transfer; Oseen estimates supply the perturbative transport; Hilbert packing supplies a square-summability mechanism for genuinely new normal defects; and null geometry isolates the branches where the quadratic term can be invisible. The next analytic task is to prove that these formal branches exhaust the localized PDE behaviour at critical scale, with all packet and component source channels closed before the final global cost statement is invoked.

A Detailed Audit Ledger: Current Source Contracts

This appendix records the current fine-grained proof audit. The entries are not a separate proof of the missing Navier-Stokes PDE source estimates. They are the source contracts, owner decompositions, and anti-circularity firewalls that the final proof must respect. The purpose is to keep the article synchronized with the present state of the proof architecture.

A.1 L153. S1 quotient packet source contract

S1 is fixed to the quotient/certified packet route. For every innovative scale,

$$e_k = \frac{d_k - P_{V_{k-1}} d_k}{\|d_k - P_{V_{k-1}} d_k\|}, \quad m_k = A_k^* e_k,$$

and in the fixed packet frame

$$m_k = \sum_{\lambda} a_{k,\lambda} \psi_{\lambda} + \tau_k.$$

The row is split as

$$a_k = a_k^{\text{old}} + a_k^{\text{cert}} + a_k^{\text{new}} + a_k^{\text{tail}} + a_k^{\text{esc}}.$$

Only a_k^{new} belongs to S1. The S1 source theorem must prove

$$\sum_k \|a_k^{\text{new}}\|_{\ell^2}^2 \leq C_{\text{S1}}(\text{Envelope}, \text{Atlas}, \text{Frame}) + C_{\text{S1}} \sum_k \text{ErrPktCompat}_k + C_{\text{S1}} \sum_k \text{EscPkt}_k,$$

equivalently boundedness of

$$T_{\text{new}} c = (\langle c, a_k^{\text{new}} \rangle)_k.$$

It must also compose with the quotient representation inequality for Innov_k . Certified-old, Hilbert-old-uncertified, recurrent, tail, and escape packet pieces are not counted by S1; they are routed to S4, S5, S6, S11, S12, VarEff, escape, or gate.

Remaining burden: prove the actual quotient packet Bessel/Carleson or direct row operator bound; prove the representation inequality for actual Navier-Stokes defects; prove packet-frame stability under pressure, Leray, cutoff, gauge, carrier, force, and chart transitions; prove recurrent physical-lift/certification alternatives without using S1; prove finite-atlas and finite-variation entry; and integrate with S4, E12, and final GQ1.

A.2 L154. S1 quotient representation inequality

For a legal S1 scale,

$$\text{Innov}_k = |\langle d_k, e_k \rangle_{H_{\text{def}}}|^2.$$

The packet pairing identity must give

$$\langle d_k, e_k \rangle_{H_{\text{def}}} = \langle b_k, a_k \rangle_{\ell^2} + \rho_{\text{frame},k} + \rho_{\text{reg},k} + \rho_{\text{native},k},$$

where b_k is the registered defect coefficient vector and a_k is the fixed packet row of $A_k^* e_k$. With the owner split of a_k , all old, certified, tail, and escape pairings are zero in the fixed owner state or charged to owned repaired errors and escapes. If

$$\|b_k\|_{\ell^2} \leq M_{\text{pkt}},$$

then Cauchy-Schwarz and Young give

$$\text{Innov}_k \leq C_{\text{S1,repr}} \|a_k^{\text{new}}\|_{\ell^2}^2 + C_{\text{S1,repr}} \text{ErrS1repr}_k.$$

This closes the formal algebra behind the representation inequality, but not the PDE facts that make the pairing identity and envelope true.

Remaining burden: prove the packet pairing identity for actual registered Navier-Stokes defects; prove the packet envelope or legal escape/update/gate; prove the S1 row Bessel/Carleson theorem; prove all representation errors independently of S1 and final GQ1.

A.3 L155. S1 packet pairing and envelope contract

Inside a fixed packet epoch, the packet analysis map is

$$\text{An}_E : H_{\text{def}} \rightarrow \ell^2(\text{Frame}_E), \quad b_k = \text{An}_E(d_k),$$

and the synthesis map satisfies

$$m_k = A_k^* e_k = \text{Syn}_E(a_k) + \tau_k.$$

The pairing identity is legal only through the fixed analysis/synthesis maps, canonical registration, and owned frame/native errors:

$$\langle d_k, e_k \rangle_{H_{\text{def}}} = \langle \text{An}_E(d_k), a_k \rangle_{\ell^2} + \rho_{\text{frame},k} + \rho_{\text{reg},k} + \rho_{\text{native},k}.$$

The coefficient envelope is the explicit observable

$$\text{PktEnv}_k = \|\text{An}_E(d_k)\|_{\ell^2(\text{Frame}_E)}^2.$$

S1 may use the representation inequality only when $\text{PktEnv}_k \leq M_{\text{pkt}}^2$, or when envelope failure is charged as explicit repaired envelope error, finite update, named packet/native/ carrier escape, chart escape, or gate.

Forbidden moves: choosing b_k after the fact; deriving the envelope from finite energy at critical normalization; changing registration, frame, dual frame, or observation map after e_k ; hiding pressure, force, cutoff, carrier, localization, memory, gauge, or normal-defect mismatch in a packet Bessel term.

Remaining burden: construct actual Navier-Stokes packet charts supplying An_E , Syn_E , and canonical registration; prove envelope outcomes; prove frame residual and native mismatch source outcomes; prove finite-atlas stabilization and S1 row Bessel/Carleson.

A.4 L156. S1 row Bessel source criterion

On a fixed or merged S1 tail atlas, the certified-new row family

$$\mathcal{A}_{\text{new}} = \{a_k^{\text{new}} : \text{Innov}_k > 0\}$$

must satisfy at least one genuine row source criterion:

- (i) finite-overlap source;
- (ii) Schur or Gram source;
- (iii) Carleson packet source;
- (iv) direct row-operator source.

Equivalently, T_{new} is bounded on the fixed packet coefficient space. The proof may not infer this from Hilbert Gram-Schmidt orthogonality of e_k in H_{def} . If no allowed row source criterion holds, the legal outcomes are packet/frame/carrier escape, finite-atlas or finite-variation failure, certified update or recurrent lift, packet-envelope escape/update, gate, or an explicitly open S1 PDE theorem.

Remaining burden: prove one allowed criterion for actual Navier-Stokes certified-new rows; prove fixed packet frame, carrier sector, dictionary, and envelope stabilization; remove recurrent/Hilbert-old rows through S5/S12 without using S1; prove independent source outcomes for surrounding errors.

A.5 L157. S1 surrounding-error independence firewall

An error term near the S1 row theorem is admissible only if it has a primary owner in S2-S12, VarEff, GQ3, or a named escape; its definition excludes $\|a_k^{\text{new}}\|^2$, the unknown T_{new} norm, and final GQ1; and its source theorem does not use the S1 row estimate.

ErrPktCompat_k is legal only as packet compatibility with NewRowSize_k = $\|a_k^{\text{new}}\|^2$ removed. It may contain frame, row-tail, sector, dictionary, owner-split, certified-old, envelope, and margin errors owned by non-S1 channels. It may not contain certified-new row mass, recurrent packet mass before the recurrent alternatives are assigned, or a packet chart change chosen after e_k .

ErrS1repr_k is split into frame pairing, registration pairing, native pairing, old-pair, certified-pair, and tail-pair owners. EscPkt_k is legal only for named packet-chart failures: unavailable frame, carrier-window exit, dictionary transport failure, envelope exit, owner-split failure, recurrent state failure, margin/floor degeneration, or finite-atlas/variation failure.

Consequently the S1 row theorem plus the representation inequality is non-circular only after these surrounding terms have source outcomes independent of S1.

A.6 L158. S4 packet compatibility source contract

S4 is the source channel for the S1-independent packet compatibility error

$$\text{ErrPktCompat}_k^{\text{ind}}.$$

On a fixed or merged packet epoch,

$$\begin{aligned} \text{ErrPktCompat}_k^{\text{ind}} = & \text{FrameCompat}_k + \text{RowTailCompat}_k + \text{SectorCompat}_k + \text{DictCompat}_k \\ & + \text{OwnerSplitCompat}_k + \text{CertOldCompat}_k \\ & + \text{EnvCompat}_k + \text{MarginPktCompat}_k. \end{aligned}$$

The owner matrix routes frame and row-tail to S4/S2/S3, sector to S9/carrier, dictionary to S6-S9/S3/S2, owner split and certified-old to S5/S11/S12, envelope to the packet-envelope owner, and margin/floor to S11/E10-E12/S12. The source target is

$$\sum_{k \in B} \text{ErrPktCompat}_k^{\text{ind}} \leq C_{S4} R_{S4}(B),$$

with all repaired-cost weights fixed before E12.

Excluded from S4: NewRowSize, the unknown T_{new} norm, recurrent-new mass before the recurrent alternatives, post-innovation chart choices, S1 row Bessel, final GQ1, and final absence of bad blocks.

A.7 L159. S4 frame compatibility owner

In a fixed packet chart with frame Frame_E, maps An_E, Syn_E, dual frame, and bounds A_E, B_E, standard frame stability gives

$$\text{FrameCompat}_k^{\text{fx}} \leq C_{\text{fr}}(\text{FrameErr}_k + \text{RowTail}_k).$$

For accepted frame transitions, ‘BuildTrans=OK’ and router S2/S4 must first route packet-frame and registration pieces, giving

$$\text{FrameCompat}_k^{\text{trans}} \leq C_{\text{fr}}^{\text{trans}}(\text{PktFrameTrans}_k + \text{RegFrameTrans}_k + \text{RowTail}_k).$$

Any term proportional to $\delta_k \|a_k^{\text{new}}\|^2$, or depending on T_{new} , is not an S4 frame error. It must be absorbed by a fixed or merged frame before S1, handled by finite variation or finite update, routed to escape/gate, or left open.

The block target is

$$\sum_{k \in B} \text{FrameCompat}_k \leq C_{\text{S4,frame}} R_{\text{frame}}(B).$$

A.8 L160. S4 row-tail/native owner

In

$$m_k = A_k^* e_k = \text{Syn}_E(a_k) + \tau_k,$$

the residual τ_k has exactly four legal outcomes: exact zero, predeclared coefficient tail, registered native/transport tail, or escape/gate. If a discarded component is admissible in the fixed packet chart and is not old, certified, tail, or escape by the owner map, it belongs to a_k^{new} , not to τ_k .

The owner size is

$$\text{RowTailOwner}_k = \text{RowTailCoeff}_k + \text{RowTailNative}_k + \text{RowTailTransport}_k + \text{RowTailReg}_k.$$

The source targets are

$$\sum_{k \in B} \text{RowTailCompat}_k \leq C_{\text{S4,tail}} \sum_{k \in B} \text{RowTailOwner}_k,$$

and, when the packet envelope is legal,

$$\sum_{k \in B} \text{RowTailPair}_k \leq C_{\text{pair,tail}} \sum_{k \in B} \text{RowTailOwner}_k.$$

Native pressure, force, cutoff/localization, carrier, chart, normal, memory, and certificate tails retain their native owners.

A.9 L161. S4 sector/carrier owner

Carrier depth, enlargement width, threshold, recentering list, and sector labels are chart data fixed before e_k . Sector mismatch has exactly one legal outcome: inside fixed sector, predeclared carrier tail, accepted carrier transport, or carrier/packet/chart escape or gate. In-sector certified-new mass cannot be hidden in sector compatibility.

The sector owner is

$$\text{SectorOwner}_k = \text{CarPkt}_k + \text{CarTail}_k + \text{CarTrans}_k + \text{CarPress}_k + \text{CarLoc}_k + \text{CarTheta}_k,$$

and

$$\sum_{k \in B} \text{SectorCompat}_k \leq C_{\text{S4,sector}} \sum_{k \in B} \text{SectorOwner}_k.$$

This is an S9/carrier theorem, not a proof of S1 row Bessel or carrier summability.

A.10 L162. S4 dictionary/component owner

Packet dictionary labels are not free coordinates. They are a finite product of component charts owned by S2, S3, S6, S7, S8, S9, S11, and S12. The dictionary owner is

$$\begin{aligned} \text{DictOwner}_k &= \text{ForceDict}_k + \text{GaugeDict}_k + \text{CutLocDict}_k + \text{CarrierDict}_k \\ &\quad + \text{RegNativeDict}_k + \text{MarginDict}_k. \end{aligned}$$

Fixed dictionaries have zero dictionary loss. Accepted component-owned mismatch satisfies

$$\sum_{k \in B} \text{DictCompat}_k \leq C_{\text{S4,dict}} \sum_{k \in B} \text{DictOwner}_k.$$

Certified-new in-dictionary mass remains a_k^{new} . Nonconstant pressure/gauge transfers, cut-off changes, carrier changes, and registration/native changes keep their primary component owners. Dictionary drift cannot replace the finite-atlas/variation route.

A.11 L163. S4 owner-split state theorem

The split

$$a_k = a_k^{\text{old}} + a_k^{\text{cert}} + a_k^{\text{new}} + a_k^{\text{tail}} + a_k^{\text{esc}}$$

is a deterministic verdict of the fixed E6-E12 effective state, packet state, recurrent alternatives, and S1 firewall. Every coefficient has exactly one verdict. In a fixed owner state, $\text{OwnerSplitCompat}_k^{\text{fix}} = 0$. For accepted owner perturbations,

$$\sum_{k \in B} \text{OwnerSplitCompat}_k \leq C_{\text{S4,owner}} \sum_{k \in B} \text{OwnerSplitLedger}_k,$$

where

$$\begin{aligned} \text{OwnerSplitLedger}_k = & \text{MemOwner}_k + \text{AtomIdOwner}_k + \text{RankCoeffOwner}_k + \text{ForceOwner}_k \\ & + \text{MarginOwner}_k + \text{FloorOwner}_k + \text{RecurrentOwner}_k + \text{TailOwner}_k. \end{aligned}$$

Certified-new mass remains in a_k^{new} . Hilbert-old but uncertified components go to S5, S12, update, escape, or gate. Recurrent components must pass through the recurrent alternatives before S1.

A.12 L164. S4 certified-old physical subtraction

A certified-old verdict is not physical subtraction. To remove a component from S1, the proof must exhibit a certified predictor

$$G_A(c_k) = G_{\text{base}} + \sum_i c_{k,i} G_i$$

with identity-preserving atom match, canonical coefficients, rank and margin control, component conventions for force, gauge, cutoff, carrier, and native norms, and nonlinear defect recomputation

$$\begin{aligned} \text{Def}_{NS}(G_A(c_k)) = & \partial_t g_A - \Delta g_A \\ & + \text{div}(g_A \otimes g_A) + \nabla \pi_A. \end{aligned}$$

The certified-old ledger is

$$\begin{aligned} \text{CertOldLedger}_k = & \text{CertAtomId}_k + \text{CertCoeffRank}_k + \text{CertDefUse}_k + \text{CertForceUse}_k \\ & + \text{CertGaugePair}_k + \text{CertCutLoc}_k + \text{CertCarrier}_k \\ & + \text{CertNativeTail}_k + \text{CertMarginFloor}_k. \end{aligned}$$

Legal certified-old epochs satisfy

$$\sum_{k \in B} \text{CertOldCompat}_k \leq C_{\text{S4,cert}} \sum_{k \in B} \text{CertOldLedger}_k.$$

Hilbert-old is not certified-old subtraction; forced-use cost is paid per use; and nonconstant pressure/gauge transfers use their registered pressure-force owner.

A.13 L165. S4 packet-envelope owner

The packet coefficient envelope

$$\text{PktEnv}_k = \|\text{An}_E(d_k)\|_{\ell^2(\text{Frame}_E)}^2$$

has exactly four legal outcomes:

- (i) bounded chart $\text{PktEnv}_k \leq M_{\text{pkt}}^2$;
- (ii) explicit repaired error $\text{ErrPktEnv}_k = \text{PktEnv}_k$ with pre-E12 weight;
- (iii) finite envelope update;
- (iv) envelope escape or gate.

The envelope owner is

$$\text{EnvOwner}_k = \text{EnvFrame}_k + \text{EnvNative}_k + \text{EnvCarrier}_k + \text{EnvMargin}_k + \text{EnvUpdate}_k,$$

and legal envelope excess satisfies

$$\sum_{k \in B} \text{EnvCompat}_k \leq C_{\text{S4}, \text{env}} \sum_{k \in B} \text{EnvOwner}_k.$$

M_{pkt} is chart data fixed before e_k , S1, and E12. Envelope failure cannot hide certified-new row mass, and ErrS1repr may use the envelope only after this owner theorem supplies a legal outcome.

A.14 L166. S4 margin and floor owner

The margin owner is

$$\begin{aligned} \text{MarginOwner}_k = & \text{ShrinkOwner}_k + \text{GraphMarginOwner}_k \\ & + \text{CertMarginOwner}_k + \text{PacketEnvMarginOwner}_k \\ & + \text{FloorOwner}_k + \text{BoundaryOwner}_k. \end{aligned}$$

MarginPktCompat has exactly four legal outcomes: fixed positive margin, registered margin comparison, finite update/floor recompute, or escape/gate. In a fixed positive chart, $\text{MarginPktCompat}_k^{\text{fix}} = 0$. For accepted margin comparisons,

$$\sum_{k \in B} \text{MarginPktCompat}_k \leq C_{\text{S4}, \text{margin}} \sum_{k \in B} \text{MarginOwner}_k.$$

Shrink stacks, graph/certificate thresholds, packet-envelope thresholds, floor sources, block lengths, and boundary data are fixed before S1/S4 use. If any change, the old epoch ends unless an explicit transport theorem preserves it with registered loss.

A.15 L167. S4 owner aggregation firewall

The aggregate

$$R_{\text{S4}}(B) = \sum_{k \in B} (R_{\text{frame},k} + R_{\text{tail},k} + R_{\text{sector},k} + R_{\text{dict},k} + R_{\text{owner},k} + R_{\text{cert},k} + R_{\text{env},k} + R_{\text{margin},k})$$

is only shorthand for eight owner ledgers. It is not an independent source theorem. On finite legal blocks, the eight local inequalities add to

$$\sum_{k \in B} \text{ErrPktCompat}_k^{\text{ind}} \leq C_{\text{S4}} R_{\text{S4}}(B).$$

On an infinite no-gate no-escape tail, S4 summability follows only if all eight owner families have summable source outcomes after finitely many certified updates/merges. If any owner exits by escape or gate, the route exits. If any owner remains an open PDE theorem, S4 remains conditional and cannot be used as a closed input to S1 or E12.

A.16 L168. S1 representation and packet escape source contract

The independent representation error is

$$\begin{aligned} \text{ErrS1repr}_k^{\text{ind}} &= \text{ErrPairFrame}_k + \text{ErrPairReg}_k + \text{ErrPairNative}_k \\ &\quad + \text{ErrOldPair}_k + \text{ErrCertPair}_k + \text{ErrTailPair}_k + \text{ErrPairEnv}_k. \end{aligned}$$

The packet escape split is

$$\begin{aligned} \text{EscPkt}_k &= \text{EscFrame}_k + \text{EscCarrier}_k + \text{EscDict}_k + \text{EscEnvelope}_k \\ &\quad + \text{EscOwnerSplit}_k + \text{EscRecurrent}_k + \text{EscMarginFloor}_k + \text{EscAtlasVar}_k. \end{aligned}$$

Representation owners must be supplied independently of S1 by S2-S12, S4, S5, and packet-envelope owners. Packet escape is legal on the S1 right-hand side only when it is registered as a summable repaired escape with pre-E12 weight. A non-summable named escape exits the no-escape branch; it is not a finite S1 error.

The combined S1 surrounding source is

$$R_{\text{sur}} = R_{\text{S4}} + R_{\text{repr}} + R_{\text{esc}},$$

where R_{S4} is legal only through L167, R_{repr} through this contract, and R_{esc} only for registered summable packet escapes.

A.17 L169. S1 ErrPairFrame owner

The first representation-error owner is

$$\text{ErrPairFrame}_k = |\rho_{\text{frame},k}|^2.$$

It is the fixed-frame pairing discrepancy after registration and native discrepancies have been removed. It is generated only by fixed analysis/synthesis duality defects, dual-frame comparison, row residual τ_k , accepted packet-frame transport, or accepted registration/frame pullback comparison.

On a bounded-envelope frame epoch,

$$\sum_{k \in B} \text{ErrPairFrame}_k \leq C_{\text{S1,pair,frame}} R_{\text{pair,frame}}(B),$$

Here $R_{\text{pair,frame}}$ is the M_{pkt}^2 -weighted sum of the frame-error, row-tail, packet-frame transport, registration-frame transport, and finite frame-update costs. The estimate is legal only after the packet-envelope owner has supplied bounded chart, repaired envelope, finite update, escape, gate, or open-PDE outcome.

This owner forbids proving frame pairing by S1 row Bessel, final GQ1, the unknown T_{new} norm, or by selecting the frame after e_k is known. Missing envelope or moving frame exits through update, envelope owner, frame escape, gate, or open theorem.

A.18 S7 pressure/gauge source architecture

The S7 pressure/gauge channel is now decomposed into five subowners:

$$\text{GaugeObs}_k = \text{GaugeGrad}_k + \text{GaugePress}_k + \text{GaugeCut}_k + \text{GaugeShrink}_k + \text{GaugePair}_k.$$

The exact zero-cost fibre is

$$\text{Gauge}_0 = \{c(t)\}.$$

Every nonconstant harmonic pressure transfer is a pressure-force pair morphism, not a gauge identity. In the notation of the audit, the pair defect is

$$\rho_k^{\text{press-force}} = \|\nabla\psi_k\|_{Y_{\text{force}}} + \text{PressTailSize}(\psi_k) + \text{GaugeLoss}_k + \text{CutoffPressureLoss}_k.$$

The local modulus is closed only in the direction

$$\text{Target}_{\text{gauge}}(B) \leq C_{\text{gauge}} \text{Obs}_{\text{gauge}}(B).$$

This is not a source theorem. The opposite direction needed for block selection is provided only by component ledgers satisfying the 243 production hypothesis.

A.19 GaugePress conditional source assembly

The pressure-tail/envelope/oscillation subchannel GaugePress is conditionally assembled from five blocks:

GP1. Pressure-envelope source closure. The harmonic envelope $PEnv$ must enter an eventual bounded pressure chart, have a produced envelope ledger, be finitely updated, enter finite variation/Cauchy/finite atlas, pay non-summable PressEsc, open a gate, or be impossible.

GP2. Pressure representative variation. The canonical representative is fixed only inside a fixed physical chart. Across chart changes, pressure representatives must have a source ledger, a source-closed owner split, finite update, finite variation/Cauchy/finite atlas, non-summable escape, gate, or impossibility.

GP3. Pressure-tail block production. On every legal pressure block,

$$\text{Obs}_{\text{GaugePress}}(B) \leq \Phi_{\text{press}}(R_{\text{press}}(B)), \quad Q_B^{\text{rep}} \geq w_{\text{press}} R_{\text{press}}(B),$$

or the pressure tail is assigned by a stored finite split to source-closed owners. If a block supremum is used, GP3 includes the required sup-to-ledger estimate.

GP4. Pressure shrink and margin import. Bounded harmonic envelope may be used through inner shrink only after θ_P , s_P , μ_P , μ_{gauge} , the E10 shrink stack, the E11 margin state, the block length, and the E12 floor data are fixed, source-controlled, finitely updated, escaped, gated, or impossible.

GP5. Cross-channel owner import. Any pressure piece crossing S7, S8, the Oseen channel, S6, or S11 must have a stored disjoint owner split. In particular, the PressPair_O term needs both pressure/gauge and collar/localization owners unless it is paid as an explicit Oseen error. The GaugeForceCmp_O term may use a short force/gauge route only after pressure and cutoff subpieces are assigned.

These five hypotheses imply the following conditional source theorem. If GP1–GP5 hold with weights, tolerances, shrink data, square/unsquare conventions, and owner splits fixed before E12, then GaugePress has exactly one of seven legal outcomes:

- (i) zero spatial-gauge pressure;
- (ii) a summable primary GaugePress ledger;

- (iii) a stored split to source-closed owners;
- (iv) finite update or finite-atlas merge;
- (v) named non-summable escape;
- (vi) gate;
- (vii) impossibility by fixed-chart/source legality.

Thus GaugePress is conditionally ready for S7 and 265, but it is not yet a nonconditional source theorem. The only unconditional piece is $c(t)$.

A.20 S7/S8/Oseen owner firewall

Oseen production is now registered on legal blocks:

$$\text{Obs}_O(B) \leq \Phi_O(R_O(B)), \quad Q_B^{\text{rep}} \geq w_O R_O(B),$$

where

$$R_O = \text{InitO} + \text{Cut}_O + \text{DriftCollar}_O + \text{PressPair}_O \\ + \text{Force}_O + \text{GaugeForceCmp}_O + \text{LocCarCut}_O.$$

This production theorem is not source summability of the Oseen owners. The pressure pieces inside R_O may be imported only after the receiving S7, S8, S6, S11, or explicit Oseen-error source theorem is closed.

Likewise, S8 cutoff/localization readiness is not S8 source closure. The pressure and collar pieces used by GaugePress, PressPair_O, GaugeForceCmp_O, and LocCarCut_O must be summable, finitely updated, non-summably escaped, gated, or impossible in their own owners. No pressure tail, cutoff pressure loss, shrink/margin loss, or pressure-force pair mass may be counted twice.

A.21 Current unresolved proof obligations

The detailed ledger above leaves the following genuinely analytic tasks:

- (i) prove source outcomes for the frame, tail, transport, carrier, dictionary, owner-split, certified-old, envelope, margin, and floor ledgers appearing in S4;
- (ii) prove the six remaining ErrS1repr owners: registration, native, old-pair, certified-pair, tail-pair, and envelope-pair;
- (iii) prove all eight EscPkt components have registered summable, finite-update, named-exit, gate, or eliminated open-PDE outcomes;
- (iv) prove actual packet pairing identity and envelope outcomes for Navier-Stokes defects;
- (v) prove true S1 row Bessel/Carleson or direct row-operator boundedness;
- (vi) prove recurrent and Hilbert-old alternatives without using S1;
- (vii) prove finite-atlas and finite-variation closure through S12/GQ3/ VarEff;
- (viii) prove E12 exact floor positivity and compatibility with envelopes, margins, updates, and block selection;
- (ix) prove GP1–GP5 and then insert the closed GaugePress theorem into full S7;

- (x) prove the remaining S7 source subchannels: gauge-gradient, gauge-cutoff, gauge-shrink, and gauge-pair, including the shared pressure-force-pair source theorem;
- (xi) prove S8 pressure/collar source closure, S10/Oseen owner source closure, the S11–E10–E11–E12 shrink-margin-floor source closure, and S12/GQ3 owner stabilization;
- (xii) close predictor differentiability, global residual representation, packet-level null-graph rigidity, carrier localization, and the quantitative closure alternative in the concrete Navier-Stokes topology;
- (xiii) assemble final GQ1 without using any source theorem after it has already been consumed.

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